

Formalizing informal mathematics — main results

Models recover an Equational Theories Project implication from natural language into a rigid grammar (ASSUME: / ASK:), graded syntactically. *Fuzzy* language is a themed Storyform narrative; *abstracted* language is a structured literal description of the same pair. Vacuous laws (E1/E2) are excluded.

informalizing-etp experiments 02, 05–08, 11 · seed 0 · OpenRouter

Story formalization needs thinking

0–57% → 93–100%

Exp 02 · 8 models, n=30 · off → on (gpt-4o-mini 70% on)

Abstracted vs fuzzy NL (no-think)

67% vs 50%

Exp 07 · mean correct, 105 pairs · literal vs story

Two-stage vs one-shot story

66% vs 50%

Exp 07 · abstract, then formalize · same 105 pairs

Majority voting (story, no-think)

26.1% → 27.8%

Exp 11 · 4 open models · temp-0 vs vote@7

1–2 · FORMALIZATION REQUIRES THINKING

With thinking on, open-weight and closed models look alike — seven of eight sit at 93–100% on the same 30 stories. With thinking off, the field spreads from 0% to 57%, and the frontier stays high: GPT-5.5 scores 92% on stories with reasoning fully off (exp 06/07), where smaller models fall into the teens and thirties.

model	think off	think on
deepseek-chat-v3.1	56.7	100
gemini-2.5-flash	56.7	100
claude-haiku-4.5	16.7	100
gpt-5-mini	16.7	100
qwen3-32b	3.3	100
llama-3.3-70b	16.7	96.7
mistral-small	6.7	93.3
gpt-4o-mini	0.0	70.0
gpt-5.5 (exp 06, strat. 24)	87.5	100

Story → rigid grammar, correct%. Rows 1–8: exp 02, uniform n=30. GPT-5.5: exp 06, non-vacuous stratified pairs.

3–4 · ABSTRACTED LANGUAGE, AND A TWO-STAGE PIPELINE

Models are far better at formalizing abstracted natural language than fuzzy narrative when they cannot think. Routing the story through an explicit abstraction step, then formalizing that description, captures most of that gap: two-stage matches the literal arm and beats one-shot story by ~16 mean points. Prompting the model to “abstract first” in a single call does not (exp 08: +0.5 / –3.2).

model	fuzzy (story)	abstracted	two-stage
gpt-5.5	92.4	99.0	99.0
claude-haiku-4.5	48.6	82.9	81.9
gemini-2.5-flash	75.2	84.8	74.3
deepseek-chat-v3.1	75.2	73.3	74.3
llama-3.3-70b	37.1	61.0	63.8
mistral-small	23.8	57.1	58.1
gpt-5-mini	45.7	55.2	58.1
qwen3-32b	35.2	52.4	50.5
gpt-4o-mini	16.2	36.2	29.5
mean	49.9	66.9	65.5

Exp 07, no-think, 105 non-vacuous pairs. Abstracted = structured literal. Two-stage = story → literal description → grammar.

5 · MAJORITY VOTING DOES NOT IMPROVE FORMALIZATION

K-sample self-consistency on the hard story arm (temperature 0.7, no thinking) does not recover accuracy. Four open-weight models move from 26.1% at a single greedy pass to 27.8% at vote@7 — a +1.7 point change at 7× cost. Errors are stable misreadings, not sampling noise around a correct modal answer.

model	temp 0	vote@1	vote@3	vote@5	vote@7
deepseek-chat-v3.1	36.5	34.0	37.0	37.5	37.0
llama-3.3-70b	25.5	25.5	26.0	27.0	27.5
mistral-small	21.5	22.0	23.5	22.5	23.0
qwen3-32b	21.0	20.5	22.5	22.5	23.5
pooled	26.1	25.5	27.2	27.4	27.8

Exp 11 · 200 synthetic pairs, story arm, reasoning off. Ballots pooled by grading-equivalence class (side-swap / dual count as the same vote).

Grading accepts variable renaming, side-swap, and consistent dualization. Thinking on/off is a uniform prompt wrapper plus the native reasoning toggle.